3.D.B.H. **LEHENENGO MAILAKO EKUAZIOAK**

Ebatzi ondorengo ekuazioak:

1. $4(x-1)-15=3(x-1)$ Soluzioa: x=16

2. $3(2-3x)-4(x-5)=6x-25$ Soluzioa: $x=\frac{51}{19}$

3. $(x-13)=4[3x-4(x-2)]$ Soluzioa: x = 9

4. $\frac{3x}{4}-\frac{x}{7}=73-2x$ Soluzioa: x= 28

5. $2x-1-\frac{3x-1}{3}-\frac{5}{3}=\frac{x+2}{6}+x-3$ Soluzioa: x = 2

6. $\frac{3(x-1)}{7}-\frac{4(x+3)}{5}=2(x+2)-7$ Soluzioa: $x=\frac{6}{83}$

7. $\frac{2(x-1)}{3}-\frac{3(x-2)}{4}=\frac{4(x-3)}{5}-\frac{3}{10}$ Soluzioa: x = 4

8. $2(x-1)-\frac{3(x+1)}{6}=6(x+2)+9x-\frac{7}{2}$ Soluzioa: $x=\frac{-22}{27}$

9. $\frac{3x-17}{8}-\frac{1-4x}{12}=\frac{1-x}{4}-\frac{9+x}{6}$ Soluzioa: $x=\frac{23}{27}$

10. $\frac{1}{8}(x-2)-\frac{2}{3}(2x+6)+x=-4$ Soluzioa: $x=\frac{-6}{5}$

11. $(x-3)^2-(x+1)(x-1)=3$ Soluzioa: $x=\frac{7}{6}$

12. $(x-1)^2-(x+1)^2=2x(x-3)-2x^2+2$ Soluzioa: x= 1

13. $(x-2)^2-(x-1)(x+1)-\frac{x-2}{3}=\frac{1}{2}(x+2)$ Soluzioa: $x=\frac{28}{29}$

14. $\frac{(2x-1)^2}{4}-\frac{(3x+1)^2}{9}=\frac{1}{6}$ Soluzioa: $x=\cancel{\frac{23}{27}}$ $-\frac{1}{60}$

3.D.B.H. LEHENENGO MAILAKO EKUAZIOAK(2)

Ebatzi ondorengo ekuazioak:

15. $(x-2)^2+3=(x-1)(-2+x)$ Soluzioa $x=5$

16. $-\frac{x+3}{2}+3(1-2x)=\frac{2(1-x)}{3}+5$ Soluzioa $x=-\frac{5}{7}$

17. $3\left(x-\frac{2}{3}\right)+4(2x-1)=\frac{x-4}{7}-2(x+4)$ Soluzioa $x=-\frac{2}{10}$

18. $\frac{4x-2}{3}-\frac{2x+4}{4}=4x+\frac{x-1}{6}$ Soluzioa $x=-\frac{9}{20}$

19. $\frac{7x^2-9}{7}-\frac{(x-3)^2}{2}=\frac{(x-2)^2}{2}-x$ Soluzioa $x=\frac{109}{84}$

20. $(2x-1)^2-(3x-4)^2=1-5x^2$ Soluzioa $x=+\frac{4}{5}$

21. $\frac{1}{8}(x-2)-\frac{2}{3}(2x+6)+x=-4$ Soluzioa $x=-\frac{6}{5}$

22. $\frac{x-2}{3}-\frac{x-3}{2}=\frac{4-2x}{5}$ Soluzioa $x=-\frac{1}{7}$

23. $\frac{(x-1)^2}{2}-\frac{(2x+1)^2}{3}=\frac{(x+2)\cdot(x-2)}{2}-\frac{4x^2}{3}$ Soluzioa $x=\frac{13}{14}$

3. DBH LEHENENGO MAILAKO EKUAZIOAK.

Ebatzi ondorengo ekuazioak:

1) $4(x-1)-15=3(x-1)$

$4x-4-15=3x-3$

$4x-3x=-3+4+15$

$\boxed{x=16}$

Egiaztapena: $4(16-1)-15=3(16-1)$

$4.15-15=3.15$

$60-15=45$

$45=45$

2) $3(2-3x)-4(x-5)=6x-25$

$6-9x-4x+20=6x-25$

$-9x-4x-6x=-25-6-20$

$-19x=-51$

$x=\frac{-51}{-19}$

$\boxed{x=\frac{51}{19}}$

3) $(x-13)=4\left[3x-4(x-2)\right]$

$x-13=4\left[3x-4x+8\right]$

$x-13=12x-16x+32$

$x-12x+16x=32+13$

$5x=45$

$x=\frac{45}{5}$

$\boxed{x=9}$

Egiaztapena: $(9-13)=4\left[3.9-4(9-2)\right]$

$-4=4.\left[27-4.7\right]$

$-4=4.\left[27-28\right]$

$-4=4.(-1)$

$-4=-4$

4) $\frac{3x}{4}-\frac{x}{7}=73-2x$

mkt(4,7)=28

$\frac{21x}{28}-\frac{4x}{28}=\frac{2044}{28}-\frac{56x}{28}$

$21x-4x=2044-56x$

$21x-4x+56x=2044$

$73x=2044$

$x=\frac{2044}{73}$

$\boxed{x=28}$

Egiaztapena: $\frac{3.28}{4}-\frac{28}{7}=73-2.28$

$21-4=73-56$

$17=17$

5) $2x-1-\frac{3x-1}{3}-\frac{5}{3}=\frac{x+2}{6}+x-3$

$\frac{12x}{6}-\frac{6}{6}-\frac{6x-2}{6}-\frac{10}{6}=\frac{x+2}{6}+\frac{6x}{6}-\frac{18}{6}$

$12x-6-(6x-2)-10=x+2+6x-18$

$12x-6-6x+2-10=x+2+6x-18$

$12x-6x-x-6x=\cancel{2}-18+6-\cancel{2}+10$

$-x=-2$

$\boxed{x=2}$

Egiaztapena:

$2.2-1-\frac{3.2-1}{3}-\frac{5}{3}=\frac{2+2}{6}+2-3$

$4-1-\frac{6-1}{3}-\frac{5}{3}=\frac{4}{6}+2-3$

$3-\frac{5}{3}-\frac{5}{3}=\frac{4}{6}+2-3$

$\frac{9-5-5}{3}=\frac{4+12-18}{6}$

$-\frac{1}{3}=\frac{-2}{6}$

$-\frac{1}{3}=\frac{-1}{3}$

6) $\frac{3(x-1)}{7}-\frac{4(x+3)}{5}=2(x+2)-7$

mkt(7,5)=35

$\frac{5.3.(x-1)-7.4.(x+3)}{35}=2x+4-7$

$15(x-1)-28(x+3)=35[2x+4-7]$

$15x-15-28x-84=70x+140-245$

$15x-28x-70x=140-245+15+84$

$-83x=-6$

$x=\frac{-6}{-83}$

$\boxed{x=\frac{6}{83}}$

7) $\frac{2(x-1)}{3}-\frac{3(x-2)}{4}=\frac{4(x-3)}{5}-\frac{3}{10}$

$\frac{8(x-1)-9(x-2)}{12}=\frac{8(x-3)-3}{10}$

$10[8(x-1)-9(x-2)]=12[8(x-3)-3]$

$10[8x-8-9x+18]=12[8x-24-3]$

$10[-x+10]=12[8x-27]$

$-10x+100=96x-324$

$-10x-96x=-324-100$

$-106x=-424$

$x=\frac{-424}{-106}\rightarrow\boxed{x=4}$

Egiaztapena:

$\frac{2(4-1)}{3}-\frac{3(4-2)}{4}=\frac{4(4-3)}{5}-\frac{3}{10}$

$\frac{2.\cancel{3}}{\cancel{3}}-\frac{3.2}{4}=\frac{4.1}{5}-\frac{3}{10}$

$2-\frac{3}{2}=\frac{4}{5}-\frac{3}{10}$

$\frac{4-3}{2}=\frac{8-3}{10}$

$\frac{1}{2}=\frac{5}{10}$

$\frac{1}{2}=\frac{1}{2}$

8) $2(x-1) - \frac{3(x+1)}{6} = 6(x+2) + 9x - \frac{7}{2}$

$$\frac{6 \cdot 2(x-1) - 3(x+1)}{6} = \frac{2 \cdot 6(x+2) + 2 \cdot 9x - 7}{2}$$

$$2[12(x-1) - 3(x+1)] = 6[12(x+2) + 18x - 7]$$

$$2[12x - 12 - 3x - 3] = 6[12x + 24 + 18x - 7]$$

$$2 \cdot [9x - 15] = 6[30x + 17]$$

$$18x - 30 = 180x + 102$$

$$18x - 180x = 102 + 30$$

$$-162x = 132$$

$$x = \frac{132}{-162} \quad \begin{matrix}(:6)\\(:6)\end{matrix}$$

$$\boxed{x = \frac{-22}{27}}$$

9) $\frac{3x-17}{8} - \frac{1-4x}{12} = \frac{1-x}{4} - \frac{9+x}{6}$ $\qquad 23/27$

$$\frac{3(3x-17) - 2(1-4x)}{24} = \frac{6(1-x) - 4(9+x)}{24}$$

$$9x - 51 - 2 + 8x = 6 - 6x - 36 - 4x$$

$$9x + 8x + 6x + 4x = 6 - 36 + 51 + 2$$

$$27x = 23$$

$$\boxed{x = \frac{23}{27}}$$

10) $\frac{1}{8}(x-2) - \frac{2}{3}(2x+6) + x = -4$

$$\frac{3(x-2) - 16(2x+6) + 24x}{24} = \frac{-96}{24}$$

$$3x - 6 - 32x - 96 + 24x = -96$$

$$3x - 32x + 24x = -96 + 6 + 96$$

$$-5x = 6$$

$$\boxed{x = \frac{-6}{5}}$$

7/6

11) $(x-3)^2-(x+1)(x-1)=3$

$(x^2-6x+9)-(x^2-1)=3$

$\cancel{x^2}-6x+9-\cancel{x^2}+1=3$

$-6x=3-9-1$

$-6x=-7$

$\boxed{x=\frac{7}{6}}$

12) $(x-1)^2-(x+1)^2=2x(x-3)-2x^2+2$

$(x^2-2x+1)-(x^2+2x+1)=\cancel{2x^2}-6x-\cancel{2x^2}+2$

$\cancel{x^2}-2x+\cancel{1}-\cancel{x^2}-2x-\cancel{1}=-6x+2$

$-4x+6x=2$

$2x=2$

$\boxed{x=1}$

Egiaztapena:

$(1-1)^2-(1+1)^2=2.1(1-3)-2.1^2+2$

$0-2^2=2.(-2)-\cancel{2}+\cancel{2}$

$-4=-4$

13) $(x-2)^2-(x-1)(x+1)-\frac{x-2}{3}=\frac{1}{2}(x+2)$

$(x^2-4x+4)-(x^2-1)-\frac{x-2}{3}=\frac{x+2}{2}$

$\frac{6(x^2-4x+4)-6(x^2-1)-2(x-2)}{\cancel{6}}=\frac{3(x+2)}{\cancel{6}}$

$\cancel{6x^2}-24x+24-\cancel{6x^2}+6-2x+4=3x+6$

$-24x-2x-3x=\cancel{6}-24-\cancel{6}-4$

$-29x=-28$

$\boxed{x=\frac{28}{29}}$

14) $$\frac{(2x-1)^2}{4} - \frac{(3x+1)^2}{9} = \frac{1}{6}$$

$$\frac{4x^2-4x+1}{4} - \frac{9x^2+6x+1}{9} = \frac{1}{6}$$

$$\frac{9(4x^2-4x+1)-4(9x^2+6x+1)}{\cancel{36}} = \frac{6}{\cancel{36}}$$

$$\cancel{36x^2}-36x+9-\cancel{36x^2}-24x-4=6$$

$$-36x-24x = 6-9+4$$

$$-60x = 1$$

$$\boxed{x = \frac{-1}{60}}$$

15) $$(x-2)^2+3 = (x-1)(-2+x)$$

$$\cancel{x^2}-4x+4+3 = -2x+\cancel{x^2}+2-x$$

$$-4x+2x+x = 2-4-3$$

$$-x = -5$$

$$\boxed{x = 5}$$

Egzaktopraw: $(5-2)^2+3 = (5-1)(-2+5)$

$$3^2+3 = 4.3$$

$$9+3 = 12$$

$$12 = 12$$

16) $$-\frac{x+3}{2} + 3(1-2x) = \frac{2(1-x)}{3} + 5$$

$$\frac{-x-3}{2} + 3 - 6x = \frac{2-2x}{3} + 5$$

$$\frac{3(-x-3)+6\cdot 3-6\cdot 6x}{\cancel{6}} = \frac{2(2-2x)+6\cdot 5}{\cancel{6}}$$

$$-3x-9+18-36x = 4-4x+30$$

$$-3x-36x+4x = 4+30+9-18$$

$$-35x = 25$$

$$x = \frac{25}{-35}$$

$$\boxed{x = \frac{-5}{7}}$$

17) $3\left(x-\frac{2}{3}\right)+4(2x-1)=\frac{x-4}{7}-2(x+4)$

$3x-\frac{\cancel{3}.2}{\cancel{3}}+8x-4=\frac{x-4}{7}-2x-8$

$\frac{7(3x-2+8x-4)}{\cancel{7}}=\frac{x-4-14x-56}{\cancel{7}}$

$21x-14+56x-28=x-4-14x-56$

$21x+56x-x+14x=-4-56+14+28$

$90x=-18$

$x=\frac{-18}{90}$

$\boxed{x=\frac{-2}{10}}$

18) $\frac{4x-2}{3}-\frac{2x+4}{4}=4x+\frac{x-1}{6}$

$\frac{4(4x-2)-3(2x+4)}{\cancel{12}}=\frac{12.4x+2(x-1)}{\cancel{12}}$

$16x-8-6x-12=48x+2x-2$

$16x-6x-48x-2x=-2+8+12$

$-40x=18$

$x=\frac{-18}{40}$

$\boxed{x=\frac{-9}{20}}$

19) $\frac{7x^2-9}{7}-\frac{(x-3)^2}{2}=\frac{(x-2)^2}{2}-x$

$\frac{2(7x^2-9)-7(x-3)^2}{\cancel{14}}=\frac{7(x-2)^2-14x}{\cancel{14}}$

$14x^2-18-7(x^2-6x+9)=7(x^2-4x+4)-14x$

$14x^2-18-7x^2+42x-63=7x^2-28x+28-14x$

$\cancel{14x^2}-\cancel{7x^2}+42x-\cancel{7x^2}+28x+14x=28+18+63$

$84x=109$

$\boxed{x=\frac{109}{84}}$

20) $(2x-1)^2-(3x-4)^2=1-5x^2$

$4x^2-4x+1-(9x^2-24x+16)=1-5x^2$

$4x^2-4x+1-9x^2+24x-16=1-5x^2$

$\cancel{4x^2}-4x-\cancel{9x^2}+24x+\cancel{5x^2}=\cancel{1}-\cancel{1}+16$

$20x=16$

$x=\frac{16}{20}$

$\boxed{x=\frac{4}{5}}$

21) $\frac{1}{8}(x-2)-\frac{2}{3}(2x+6)+x=-4$

$\frac{3(x-2)-8\cdot 2(2x+6)+24x}{\cancel{24}}=\frac{24\cdot(-4)}{\cancel{24}}$

$3x-6-32x-96+24x=-96$

$3x-32x+24x=\cancel{-96}+6+\cancel{96}$

$-5x=6$

$\boxed{x=\frac{-6}{5}}$

22) $\frac{x-2}{3}-\frac{x-3}{2}=\frac{4-2x}{5}$

$\frac{10(x-2)-15(x-3)}{\cancel{30}}=\frac{6(4-2x)}{\cancel{30}}$

$10x-20-15x+45=24-12x$

$10x-15x+12x=24+20-45$

$7x=-1$

$\boxed{x=\frac{-1}{7}}$

23) $$\frac{(x-1)^2}{2} - \frac{(2x+1)^2}{3} = \frac{(x+2)\cdot(x-2)}{2} - \frac{4x^2}{3}$$

$$\frac{3(x-1)^2 - 2(2x+1)^2}{\cancel{6}} = \frac{3(x+2)\cdot(x-2) - 2\cdot 4x^2}{\cancel{6}}$$

$$3(x^2-2x+1) - 2(4x^2+4x+1) = 3(x^2-4) - 8x^2$$

$$3x^2 - 6x + 3 - 8x^2 - 8x - 2 = 3x^2 - 12 - 8x^2$$

$$\cancel{3x^2} - 6x - \cancel{8x^2} - 8x - \cancel{3x^2} + \cancel{8x^2} = -12 - 3 + 2$$

$$-14x = -13$$

$$x = \frac{-13}{-14}$$

$$\boxed{x = \frac{13}{14}}$$